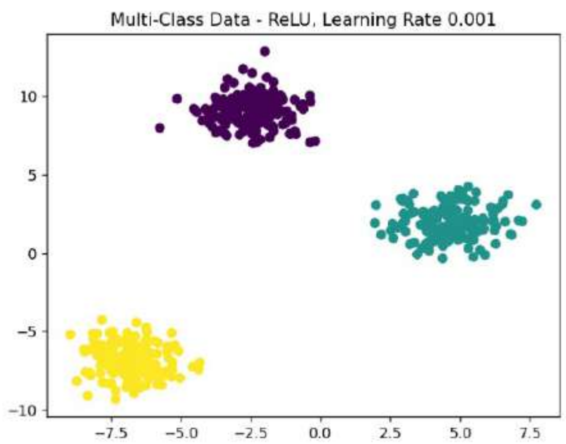
exp 28.py - C:/Users/shaik/OneDrive/Desktop/MLA0405/exp 28.py (3.13.5)

```python
import matplotlib.pyplot as plt
from sklearn.datasets import make_blobs
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score

X, y = make_blobs(
    n_samples=500, centers=3, n_features=2,
    random_state=42
)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

model = MLPClassifier(
    hidden_layer_sizes=(10,),
    activation="relu",
    learning_rate_init=0.001,
    max_iter=1000,
    random_state=42
)

model.fit(X_train, y_train)
y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))

plt.scatter(X[:, 0], X[:, 1], c=y)
plt.title("Multi-Class Data - ReLU, Learning Rate 0.001")
plt.show()
```

IDLE Shell 3.13.5

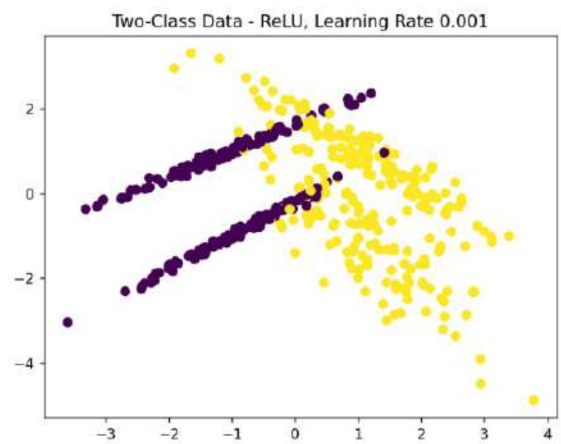
```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/Users/shaik/OneDrive/Desktop/MLA0405/exp 28.py ==========
Accuracy: 1.0
```

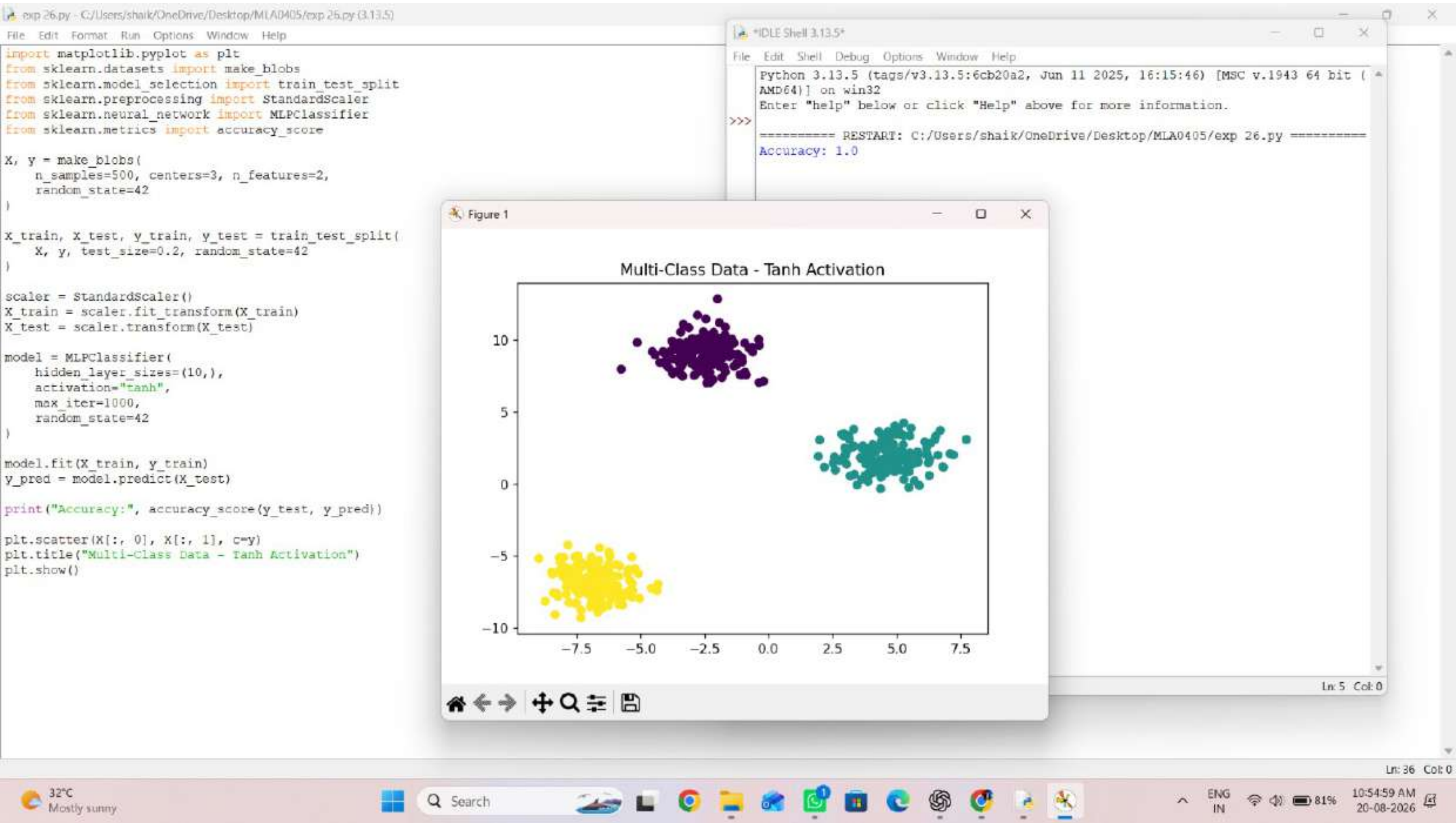
Figure 1



```python
import matplotlib.pyplot as plt
from sklearn.datasets import make_classification
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score

X, y = make_classification(
    n_samples=500, n_features=2, n_redundant=0,
    n_informative=2, n_classes=2, random_state=42
)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

model = MLPClassifier(
    hidden_layer_sizes=(10,),
    activation="relu",
    learning_rate_init=0.001,
    max_iter=1000,
    random_state=42
)

model.fit(X_train, y_train)
y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))

plt.scatter(X[:, 0], X[:, 1], c=y)
plt.title("Two-Class Data - ReLU, Learning Rate 0.001")
plt.show()
```

```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.

========== RESTART: C:/Users/shaik/OneDrive/Desktop/MLA0405/exp 27.py ==========
Accuracy: 0.89
```





exp 26.py - C:/Users/shaik/OneDrive/Desktop/MLA0405/exp 26.py (3.13.5)
File Edit Format Run Options Window Help
import matplotlib.pyplot as plt
from sklearn.datasets import make_blobs
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score
X, y = make_blobs(
n_samples=500, centers=3, n_features=2,
random_state=42
)
X_train, X_test, y_train, y_test = train_test_split(
X, y, test_size=0.2, random_state=42
)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
model = MLPClassifier(
hidden_layer_sizes=(10,),
activation="tanh",
max_iter=1000,
random_state=42
)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("Accuracy:", accuracy_score(y_test, y_pred))
plt.scatter(X[:, 0], X[:, 1], c=y)
plt.title("Multi-Class Data - Tanh Activation")
plt.show()
IDLE Shell 3.13.5
File Edit Shell Debug Options Window Help
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/Users/shaik/OneDrive/Desktop/MLA0405/exp 26.py ==========
Accuracy: 1.0
Figure 1
Multi-Class Data - Tanh Activation
10
5
0
−5
−10
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
Ln: 5 Col: 0
Ln: 36 Col: 0
32°C
Mostly sunny
Search
ENG
IN
81%
10:54:59 AM
20-08-2026

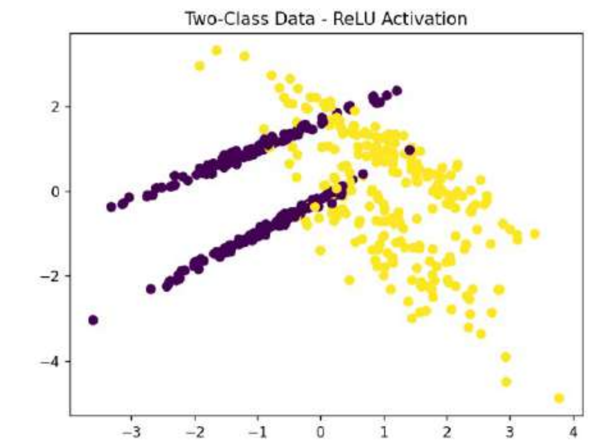
exp 22.py - C:/Users/shaik/OneDrive/Desktop/MLA0405/exp 22.py (3.13.5)

```python
import matplotlib.pyplot as plt
from sklearn.datasets import make_classification
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score

X, y = make_classification(
    n_samples=500, n_features=2, n_redundant=0,
    n_informative=2, n_classes=2, random_state=42
)

x_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

model = MLPClassifier(
    hidden_layer_sizes=(10,),
    activation="relu",
    max_iter=1000,
    random_state=42
)

model.fit(X_train, y_train)
y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))

plt.scatter(X[:, 0], X[:, 1], c=y)
plt.title("Two-Class Data - ReLU Activation")
plt.show()
```

IDLE Shell 3.13.5

```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/Users/shaik/OneDrive/Desktop/MLA0405/exp 22.py ==========
Accuracy: 0.89
```

Figure 1



exp 24.py - C:/Users/shaik/OneDrive/Desktop/MLA0405/exp 24.py (3.13.5)

```
import matplotlib.pyplot as plt
from sklearn.datasets import make_blobs
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score

X, y = make_blobs(
    n_samples=500, centers=3, n_features=2,
    cluster_std=1.5, random_state=42
)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

model = MLPClassifier(
    hidden_layer_sizes=(10,),
    activation="logistic",
    max_iter=1000,
    random_state=42
)

model.fit(X_train, y_train)
y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))

plt.scatter(X[:, 0], X[:, 1], c=y)
plt.title("Multi-Class Data - Sigmoid Activation")
plt.show()
```

IDLE Shell 3.13.5

```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/Users/shaik/OneDrive/Desktop/MLA0405/exp 24.py ==========
Accuracy: 1.0
```

Figure 1